

# SkyFinder Temperature Prediction with Deep Imbalanced Regression

*Brief take-home experiment report for the Health Intelligence Lab*

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**Main finding** Label Distribution Smoothing reduced validation MAE from 2.677 °C to 2.610 °C, but test MAE increased from 4.922 °C to 6.315 °C. The result suggests that label imbalance matters in some temperature ranges, while temporal distribution shift may be an important contributor to the generalization gap in this subset.

## Objective

I adapted the Label Distribution Smoothing component of Deep Imbalanced Regression to predict ambient temperature from SkyFinder images. The experiment compares a standard ResNet-18 regressor with the same model trained using LDS-weighted loss.

## Method

Images from cameras 858, 3888, and 4795 were joined with the SkyFinder weather metadata. Invalid temperatures and two truncated JPEG files were removed. Samples were split chronologically by capture date into 1,777 training, 375 validation, and 395 test images, which prevents adjacent frames from the same date from being divided across splits.

The backbone was an ImageNet-pretrained ResNet-18 with its classification layer replaced by a one-output regression head. Images were resized to 224 by 224 pixels and normalized with ImageNet statistics; random horizontal flipping was applied only during training. Both models used AdamW, a learning rate of 0.0001, weight decay of 0.0001, batch size 32, 20 epochs, and random seed 42. The baseline minimized mean absolute error.

For LDS, training temperatures were placed into 1 °C bins. Their counts were smoothed with a Gaussian kernel of size 5 and sigma 2. Each sample received a weight inversely proportional to the smoothed density of its temperature bin; weights were normalized to mean 1. The weighted absolute error was used only for optimization, while ordinary MAE was retained for validation and checkpoint selection.

## Experimental Results

| Model              | Val MAE | Val RMSE | Test MAE | Test RMSE |
|--------------------|---------|----------|----------|-----------|
| Training median    | 10.226  | 11.010   | 5.945    | 7.803     |
| ResNet-18 baseline | 2.677   | 3.459    | 4.922    | 6.293     |
| ResNet-18 with LDS | 2.610   | 3.312    | 6.315    | 7.551     |

Table 1 Temperature error in degrees Celsius

On validation data, LDS improved MAE by 0.066 °C (2.5 percent) and RMSE by 0.148 °C (4.3 percent), calculated from the unrounded metrics. Its largest validation gains were below 0 °C: MAE fell from 4.175 °C to 3.558 °C in the −10 to −5 °C bin and from 3.409 °C to 2.432 °C in the −5 to 0 °C bin.

## Analysis

The validation result supports the intended role of LDS: neighboring temperature labels share information, and rare low-temperature examples receive greater influence during training. The improvement was not uniform. For example, validation MAE in the 5 to 10 °C bin rose from 6.316 °C to 7.706 °C, showing that inverse-density weighting can trade performance between regions rather than improve the complete range.

The chronological test split was substantially colder than the training and validation splits. Mean temperatures were 5.89 °C for the test set, 11.36 °C for the training set, and 13.25 °C for the validation set. Both models overpredicted on the test set, but the mean prediction bias was larger with LDS (+4.95 °C) than with the baseline (+3.82 °C). LDS therefore improved validation performance but did not improve performance on the final test period. These results suggest that temporal distribution shift and limited data coverage may be important contributors to the test gap, although this conclusion is limited to the current subset and a single chronological split.

## What I Learned

This project taught me how to set up a basic ResNet regression pipeline, replace its classification head, and adjust key training parameters. I gained a clearer understanding of how convolutional layers extract spatial features from images and how those features can support continuous prediction rather than only classification. I also learned how LDS smooths neighboring label frequencies and gives underrepresented temperature ranges greater influence during training. I have long been interested in image recognition, and implementing the complete pipeline turned these ideas into practical understanding and strengthened my interest in vision-based machine learning.

## Improvements

- Tune LDS bin width, kernel size, sigma, and a maximum weight cap on the validation set; the current maximum sample weight was about 24.1.
- Implement Feature Distribution Smoothing and compare baseline, LDS, FDS, and LDS plus FDS under the same protocol.
- Add more cameras and seasons, or design seasonal splits, to reduce the train-test temperature shift while preserving temporal separation.
- Run multiple random seeds and report uncertainty; examine time of day, weather metadata, and prediction calibration as possible sources of error.

## References

1. Yang, Y., Zha, K., Chen, Y., Wang, H., and Katabi, D. Delving into Deep Imbalanced Regression. ICML, 2021. [PMLR paper](#)
2. Mihail, R. P. et al. SkyFinder dataset. [Dataset page](#)